WORKED SOLUTION

STEP I 2015, Question 9

Mechanics • Projectile Motion

STEP I 2015, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q9

The key idea

"Last to hit the ground" is about the *landing time*, not the firing time: a bullet fired later at a lower angle spends less time in the air, so it may land before an earlier one. Write the landing time as firing time plus time of flight, and maximise it over the firing period. Using the angle α as the variable instead of t makes the calculus clean, and the condition on k decides whether the maximum is at a stationary point or at an end.

Solution

Setting up

Model each bullet as a particle moving freely under gravity, with no air resistance, fired from ground level. A bullet fired with elevation α has vertical position $y = u\sin\alpha\, s - \frac{1}{2}gs^2$ at time s after firing, so it returns to the ground after

$$\frac{2u\sin\alpha}{g},$$

and in that time it travels horizontally

$$R(\alpha) = u\cos\alpha\cdot\frac{2u\sin\alpha}{g} = \frac{2u^2\sin\alpha\cos\alpha}{g}.$$

The bullet with elevation α is fired at time $t = (\frac{1}{3}\pi - \alpha)/\lambda$, and as t runs over $[0, \frac{\pi}{6\lambda}]$, α runs over $[\frac{1}{6}\pi, \frac{1}{3}\pi]$, each value once. So this bullet lands at time

$$L(\alpha) = \frac{\frac{1}{3}\pi - \alpha}{\lambda} + \frac{2u\sin\alpha}{g}, \qquad \tfrac{1}{6}\pi \le \alpha \le \tfrac{1}{3}\pi.$$

The last bullet to land is the one that maximises L. Since $k = \dfrac{g}{2\lambda u}$, we have $\dfrac{2u}{g} = \dfrac{1}{\lambda k}$, and

$$L(\alpha) = \frac{1}{\lambda}\left(\tfrac{1}{3}\pi - \alpha + \frac{\sin\alpha}{k}\right), \qquad L'(\alpha) = \frac{1}{\lambda}\left(\frac{\cos\alpha}{k} - 1\right) = \frac{\cos\alpha - k}{\lambda k}.$$

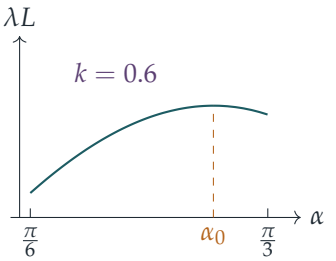
So L increases where $\cos\alpha > k$ and decreases where $\cos\alpha < k$. On our interval, $\cos\alpha$ decreases from $\frac{1}{2}\sqrt{3}$ (at $\alpha = \frac{1}{6}\pi$) to $\frac{1}{2}$ (at $\alpha = \frac{1}{3}\pi$).

Ask yourself Why does a larger α mean an *earlier* firing time, and why does that matter for the sign of L'?

The barrel is lowering, so the steep bullets go first. Raising α by a little moves the firing time earlier by $1/\lambda$ per radian, but adds $\frac{2u}{g}\cos\alpha$ per radian to the flight time. $L'(\alpha)$ is the balance of these two effects, and the sign tells us which wins.

The case $\frac{1}{2} \le k \le \frac{1}{2}\sqrt{3}$

Now k lies in the range of $\cos\alpha$, so there is exactly one $\alpha_0 \in [\frac{1}{6}\pi, \frac{1}{3}\pi]$ with $\cos\alpha_0 = k$. Because cos is decreasing on the interval, $\cos\alpha > k$ for $\alpha < \alpha_0$ and $\cos\alpha < k$ for $\alpha > \alpha_0$. So L increases up to α_0 and decreases after it, and the bullet fired at elevation α_0 is the last to land. (If $k = \frac{1}{2}$ or $k = \frac{1}{2}\sqrt{3}$, then α_0 is an endpoint and L is monotonic, but the conclusion is the same.)



(The vertical axis is shifted so that the variation in landing time is visible.)

Its range is, using $\sin\alpha_0 = \sqrt{1-k^2}$ (positive since α_0 is acute),

$$R(\alpha_0) = \frac{2u^2 \sin\alpha_0 \cos\alpha_0}{g} = \frac{2ku^2\sqrt{1-k^2}}{g},$$

which is the required distance.

The case $k < \frac{1}{2}$

Now $\cos\alpha \ge \frac{1}{2} > k$ for every α in the interval, so $L'(\alpha) > 0$ throughout and L is strictly increasing. The maximum is at $\alpha = \frac{1}{3}\pi$: the *first* bullet fired is the last to land. Its range is

$$R\left(\tfrac{1}{3}\pi\right) = \frac{2u^2}{g} \cdot \frac{\sqrt{3}}{2} \cdot \frac{1}{2} = \frac{\sqrt{3}\,u^2}{2g}.$$

So for $k < \frac{1}{2}$ the last bullet lands a distance $\dfrac{\sqrt{3}\,u^2}{2g}$ from the gun.

(For completeness: if $k > \frac{1}{2}\sqrt{3}$, then L is decreasing, the final bullet fired is the last to land, and since $\sin\frac{\pi}{3} = \sin\frac{2\pi}{3}$ the distance is again $\frac{\sqrt{3}\,u^2}{2g}$.)

Checks

- Continuity: at $k = \frac{1}{2}$ the first formula gives $\dfrac{2 \cdot \frac{1}{2}u^2 \cdot \frac{\sqrt{3}}{2}}{g} = \dfrac{\sqrt{3}\,u^2}{2g}$, agreeing with the $k < \frac{1}{2}$ answer; at $k = \frac{1}{2}\sqrt{3}$ it gives the same value again.
- Taking $u^2/g = 1$ and $k = 0.6$, the last bullet has $\cos\alpha_0 = 0.6$ and lands at distance $2(0.6)(0.8) = 0.96$. A computer check sampling twenty thousand firing times, and stepping the motion of that bullet numerically, agreed.
- Units: k is dimensionless (g over λu, which is speed per time), and u^2/g is a length.

Common slips

- Maximising the range, or the time of flight alone, instead of the landing time $t + \frac{2u\sin\alpha}{g}$.
- Assuming the last bullet fired is the last to land.
- Getting the sign of L' wrong by forgetting that t decreases as α increases.

Beyond STEP

Maximising a function over a closed interval means comparing interior stationary points with the endpoints, and here a parameter decides which one wins. This kind of "which constraint is active" switch is the starting point of constrained optimisation, which you will meet in first-year methods courses. For more mechanics in this spirit, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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